

Chapter 1

Introduction

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Objectives

- ❑ To provide a **grand tour** of the major operating systems components
- ❑ To provide **coverage** of basic computer system organization



Outlines for Chapter 1

- ❑ What Operating Systems Do
- ❑ Computer-System Organization
- ❑ Operating-System Structure
- ❑ Operating-System Operations
- ❑ Process Management
- ❑ Memory Management
- ❑ Storage Management
- ❑ Protection and Security



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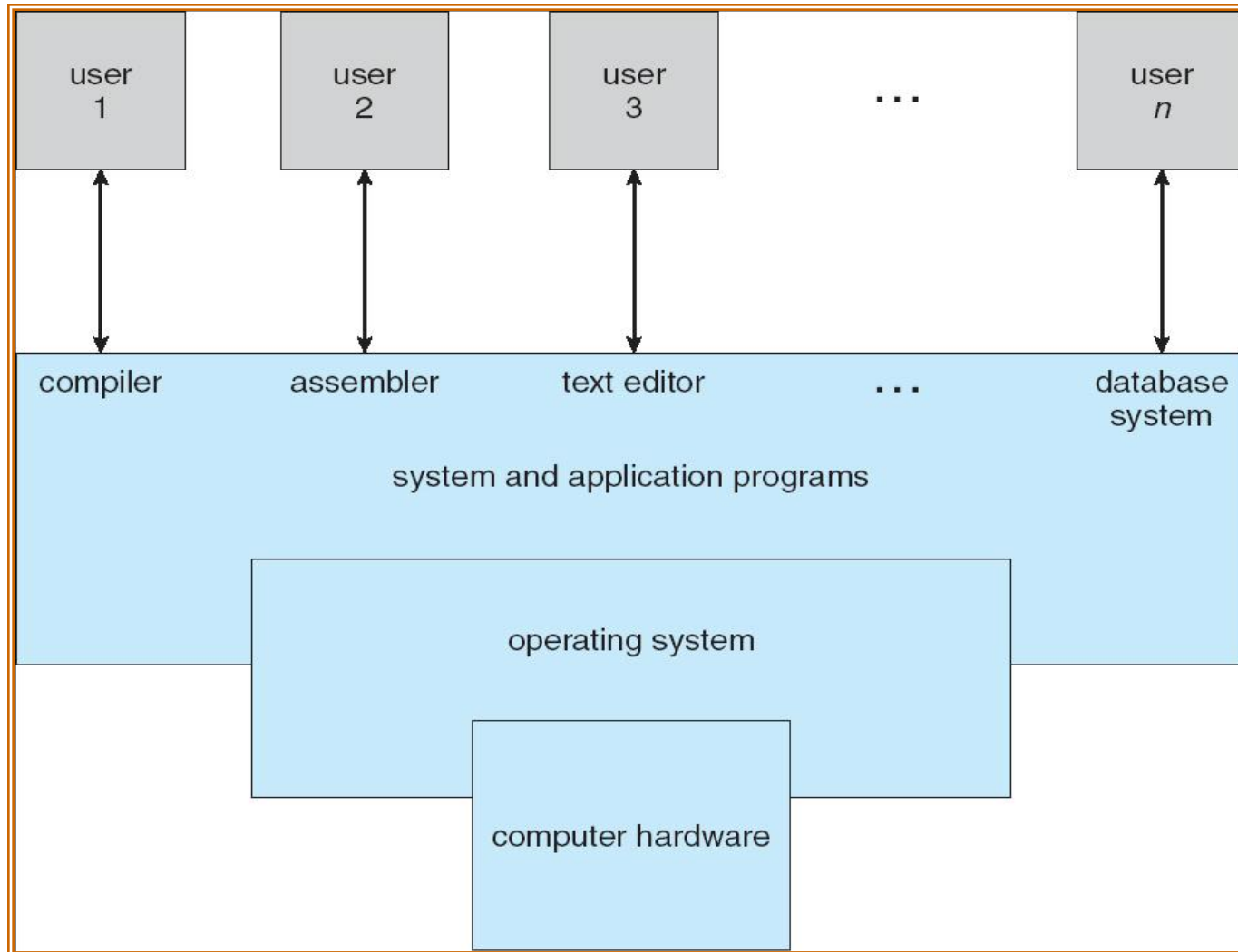


Computer System Structure

- ❑ Computer system can be divided into four components
 - **Hardware** – provides basic computing resources
 - CPU, memory, I/O devices
 - **Operating system**
 - Controls and coordinates use of hardware among various applications and users
 - **System & application programs**
 - define the ways in which the system resources are used to solve the computing problems of the users
 - word processors, compilers, web browsers, database systems, video games
 - **Users**



Four Components of a Computer System



What is an Operating System?

- ❑ A program that acts as an intermediary between a *user* of a computer and the computer *hardware*.
- ❑ Fundamental goal of computer systems
 - Execute user programs and make solving user problems easier.
- ❑ Goals of Operating System:
 - Make the computer system **convenient** to use.
 - Use the computer hardware in an **efficient** manner.



Operating System Definition

❑ OS is a **resource allocator**

- Manages all resources
- Decides between *conflicting requests* for efficient and fair resource use

❑ OS is a **control program**

- Controls execution of programs to prevent *errors* and *improper use* of the computer



Operating System Definition (Cont.)

- ❑ No universally accepted definition
- ❑ “Everything a vendor ships when you order an operating system” is good approximation
 - But varies wildly
- ❑ “The one program running at all times on the computer” is the **kernel**. Everything else is either a system program (ships with the operating system) or an application program



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Computer Startup

❑ **bootstrap program** (or bootstrap loader) is loaded at power-up or reboot

引导程序

Erasable Programmable Read-Only Memory

- Typically stored in **firmware**, generally known as **firmware**

固件，固化在硬件中的软件

- **Initialize** all aspects of system

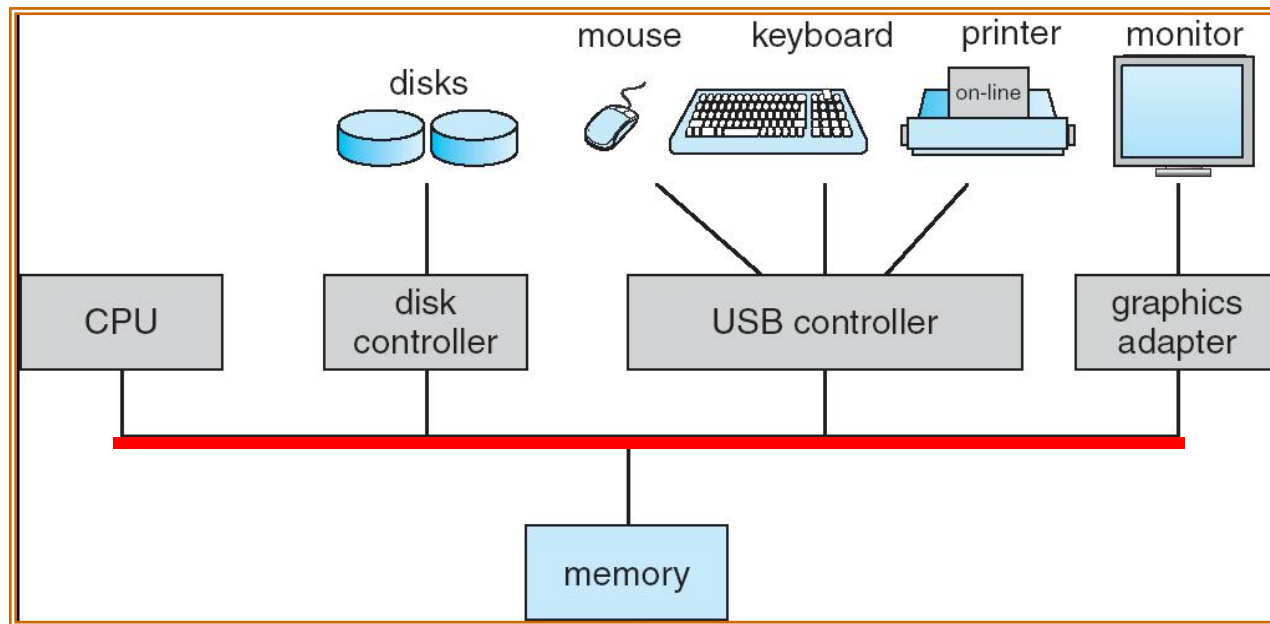
- **Loads** operating system kernel and **starts** execution



Computer System Operation

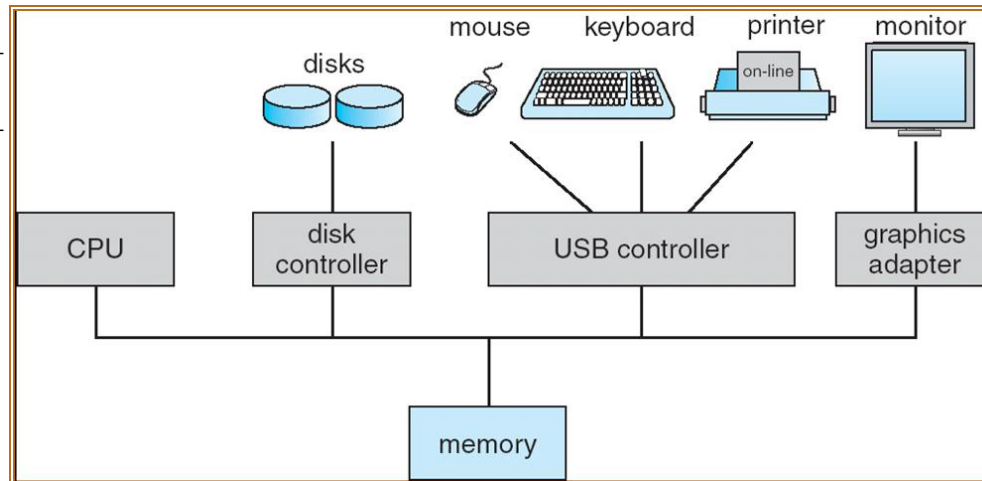
❑ Computer-system operation

- One or more CPUs, device controllers connect through *common bus* providing access to shared memory
- Concurrent execution of CPUs and devices *competing* for memory cycles



Computer-System Operation (cont'd)

- I/O devices and the CPU can execute *concurrently*.
- Each device controller has a **local buffer** (*some special purpose registers*).
- CPU moves data between *main memory* and *local buffers*
- I/O is between the device and local buffer of controller.
- Device finishes its operation and sends an *interrupt*.

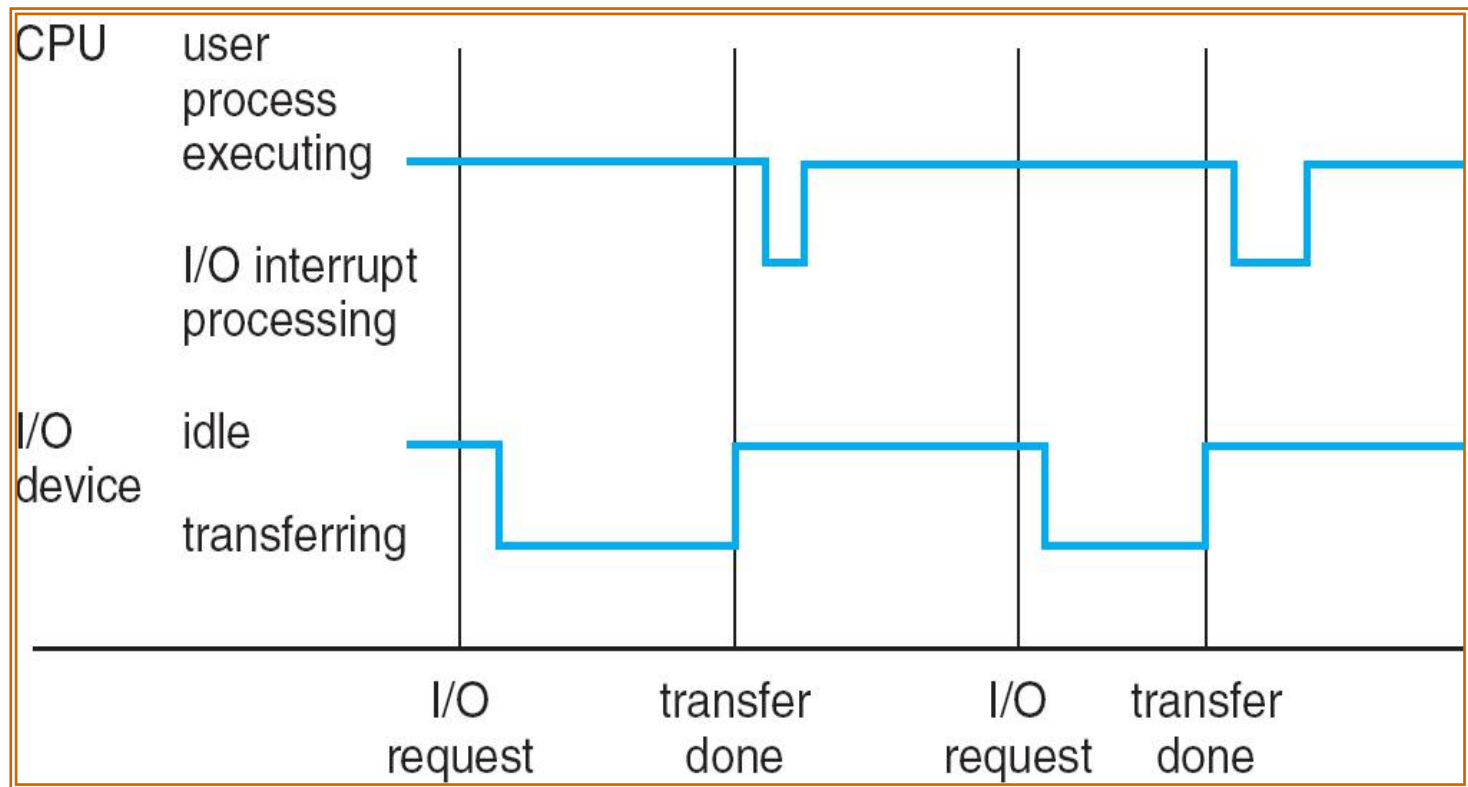


Common Functions of Interrupts

- ❑ Interrupt **transfers control** to the interrupt service routine generally, through the *interrupt vector*, which contains the addresses of all the service routines.
- ❑ Interrupt architecture must save the address of the interrupted instruction.
- ❑ A *trap* is a software-generated interrupt caused either by an error or a user request (the latter is often referred to as a *system call*).
- ❑ An operating system is interrupt driven.

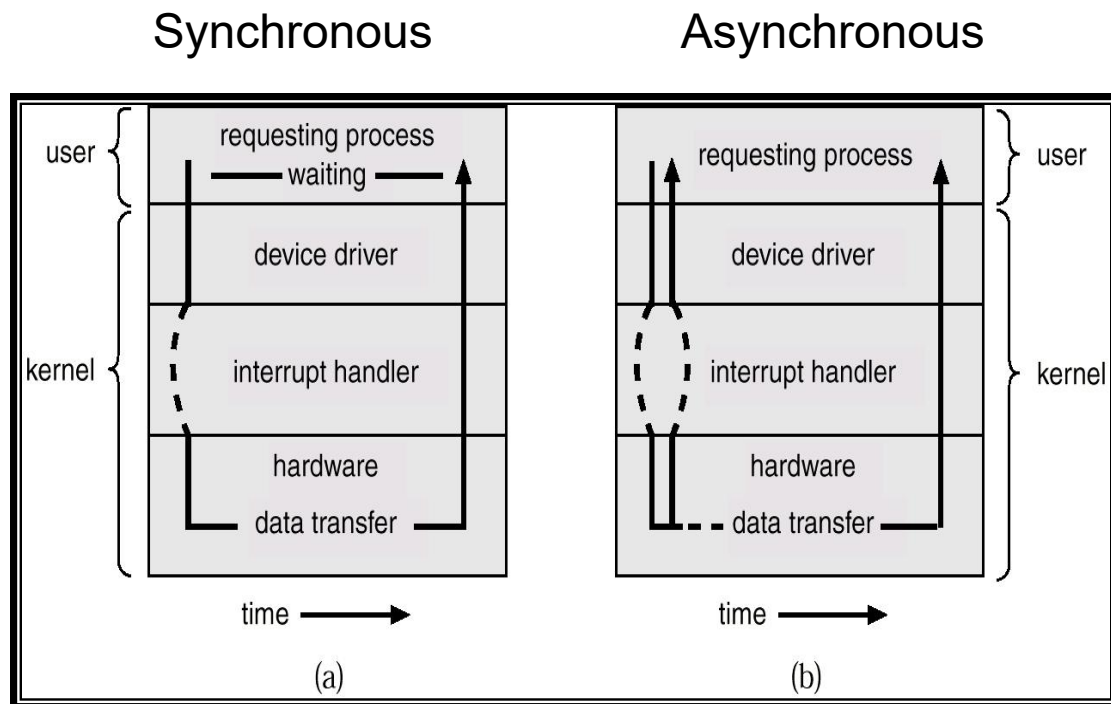


Interrupt Timeline



I/O Method

- ❑ After I/O starts, control returns to user program only upon I/O completion.
- ❑ After I/O starts, control returns to user program without waiting for I/O completion.



Direct Memory Access Structure

- ❑ Used for high-speed I/O devices able to transmit information at close to memory speeds.
- ❑ Device controller transfers blocks of data from buffer storage directly to main memory without CPU intervention.
- ❑ Only one interrupt is generated per block, rather than the one interrupt per byte.



Storage Structure

❑ Main memory

- only large storage media that the CPU can access directly.

❑ Secondary storage

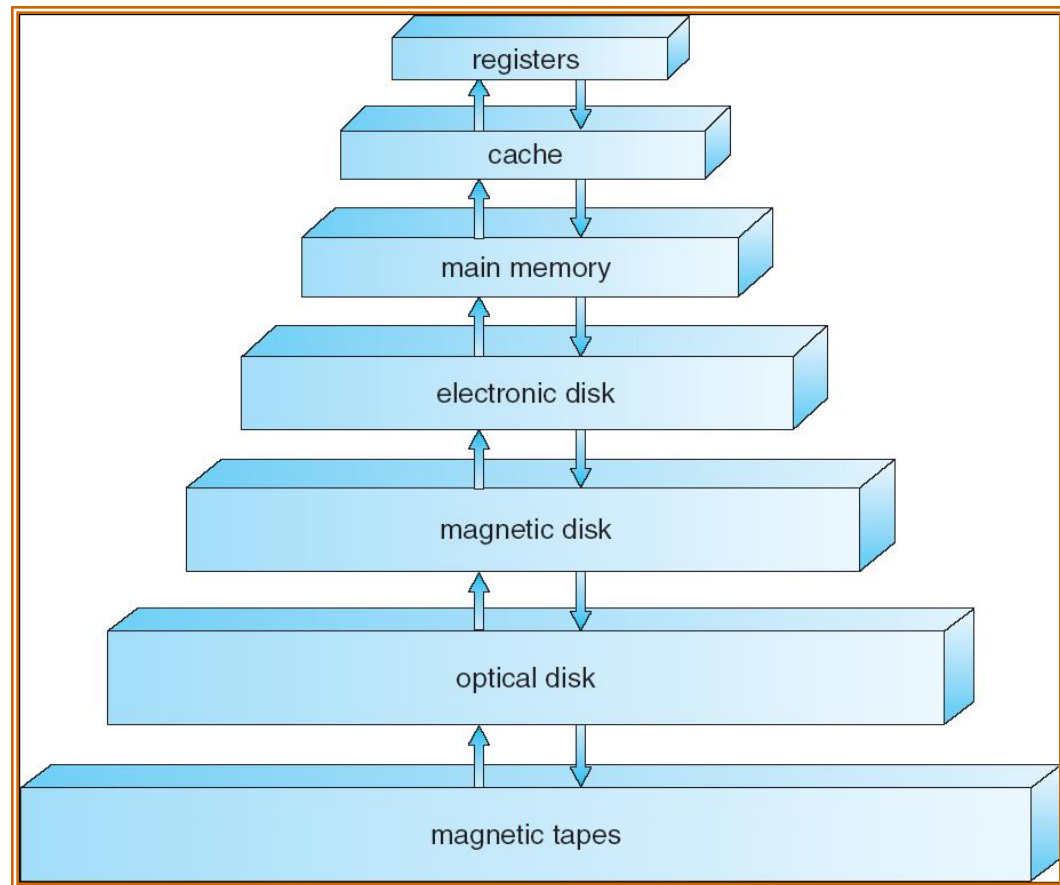
- extension of main memory that provides *large nonvolatile* storage capacity.
- Most common: Magnetic disks
 - rigid metal or glass platters covered with magnetic recording material
 - Disk surface is logically divided into *tracks*, which are subdivided into *sectors*.
 - The *disk controller* determines the logical interaction between the device and the computer.



Storage Hierarchy

❑ Storage systems organized in hierarchy.

- Speed
- Cost
- Size
- Volatility



Performance of various levels of Storage

Level	1	2	3	4
Name	registers	cache	main memory	disk storage
Typical size	< 1 KB	> 16 MB	> 16 GB	> 100 GB
Implementation technology	custom memory with multiple ports, CMOS	on-chip or off-chip CMOS SRAM	CMOS DRAM	magnetic disk
Access time (ns)	0.25 – 0.5	0.5 – 25	80 – 250	5,000.000
Bandwidth (MB/sec)	20,000 – 100,000	5000 – 10,000	1000 – 5000	20 – 150
Managed by	compiler	hardware	operating system	operating system
Backed by	cache	main memory	disk	CD or tape



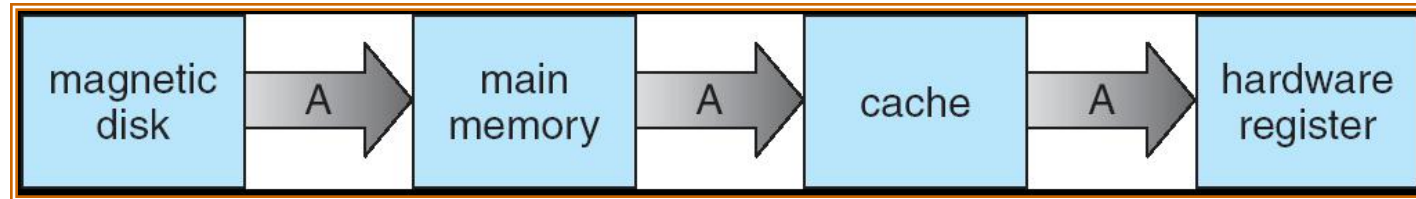
Caching

❑ *Caching*

- Copying Information in use from slower to faster storage temporarily
- ❑ Faster storage (cache) checked first to determine if information is there
 - If it is, information used directly from the cache (fast)
 - If not, use the information directly from the source, and putting a copy in the cache under the assumption that we will need it again soon
- ❑ Important principle, performed at many levels in a computer (in hardware, operating system, software)
- ❑ Cache smaller than storage being cached



Migration of Integer A from Disk to Register



- ❑ Movement between levels of storage hierarchy can be explicit or implicit
- ❑ Multitasking environments must be careful to use most recent value, no matter where it is stored in the storage hierarchy (see figure)
- ❑ Multiprocessor environment must provide cache coherency in hardware such that all CPUs have the most recent value in their cache



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Operating System Structure

❑ Multiprogramming needed for efficiency (CPU utilization)

- Single user cannot keep CPU and I/O devices busy at all times
- Multiprogramming organizes jobs so CPU always has one to execute
- A subset of total jobs in system is kept in memory
- One job selected and run via **job scheduling**
- When it has to wait (for I/O for example), OS switches to another job

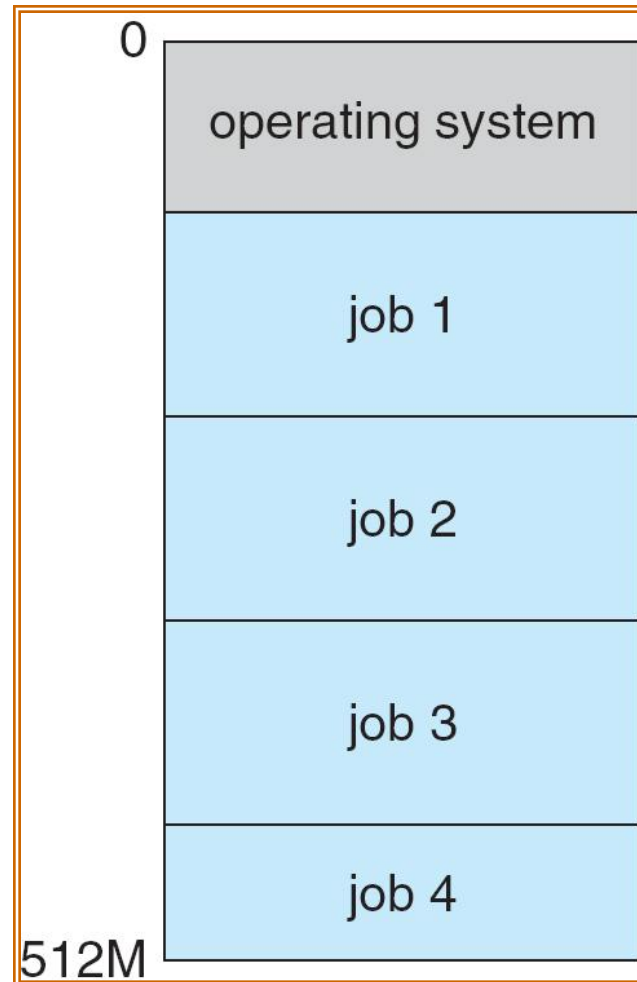


Operating System Structure

- ❑ **Timesharing** (multitasking) is logical extension, in which CPU switches jobs so frequently that users can interact with each job while it is running (**interactivity**)
 - **Response time** should be < 1 second
 - Each user has at least one program executing in memory [**process**]
 - If several jobs ready to run at the same time [**CPU scheduling**]
 - If processes don't fit in memory, **swapping** moves them in and out to run
 - **Virtual memory** allows execution of processes not completely in memory



Memory Layout for Multi-programmed System



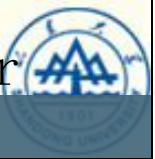
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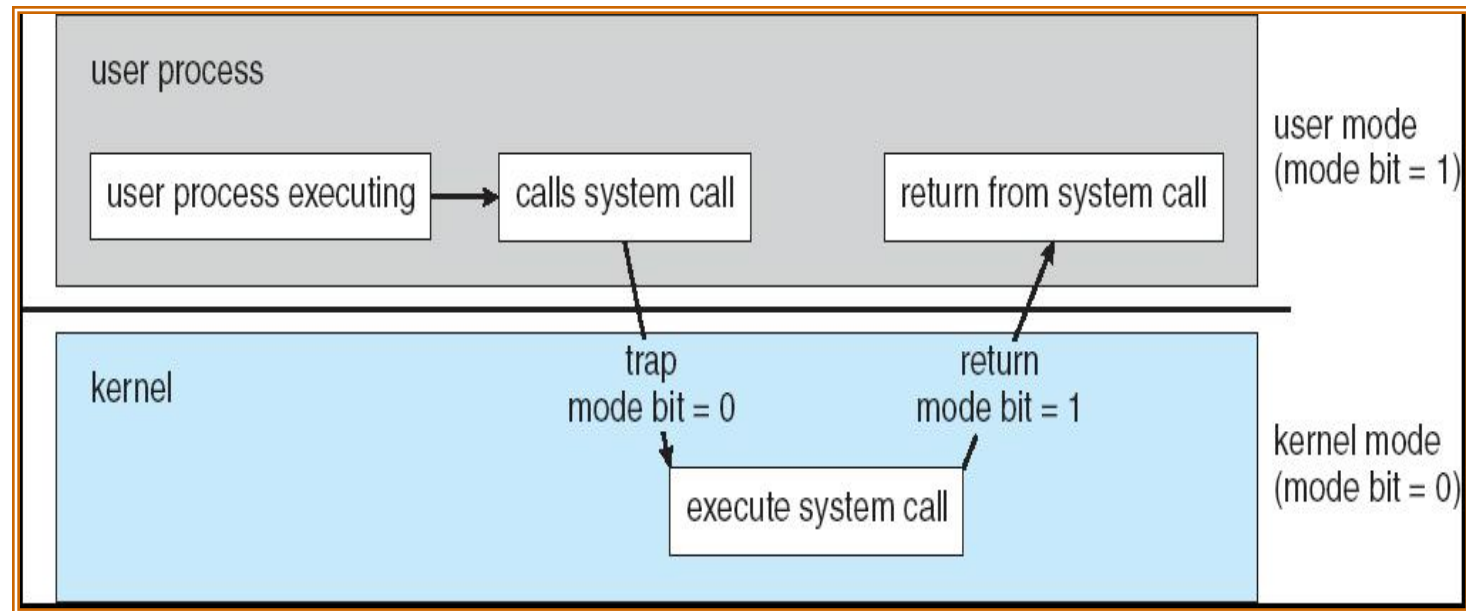


Operating-System Operations

- ❑ Modern operating systems are Interrupt driven.
- ❑ Software error or request creates **trap** (or **exception**)
 - Division by zero, request for operating system service
- ❑ Dual-mode operation allows OS to protect itself and other system components
- ❑ User mode and kernel mode,
 - **Mode bit** provided by hardware provides ability to distinguish when system is running user code or kernel code
 - Some instructions designated as **privileged**, only executable in kernel mode
 - System call changes mode to kernel, resets it to user when return from call



Transition from User to Kernel Mode



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Process Management

- ❑ A **process** is a program in execution.
It is a unit of work within the system.
- ❑ Program is a *passive* entity, process is an *active* entity.
- ❑ Process needs resources to accomplish its task
 - CPU, memory, I/O, files
- ❑ Process termination requires **reclaim** of any reusable resources



Process Management

- ❑ Typically system has many processes, some user running concurrently on one or more CPUs
 - Concurrency by multiplexing the CPUs among the processes / threads
- ❑ **Single-threaded** process has one *program counter* specifying location of next instruction to execute
 - Process executes instructions sequentially, one at a time, until completion
- ❑ **Multi-threaded** process has one program counter per thread



Process Management Activities

- ❑ The operating system is responsible for the following activities in connection with process management:
 - Creating and deleting both user and system processes
 - Suspending and resuming processes
 - Providing mechanisms for process synchronization
 - Providing mechanisms for process communication
 - Providing mechanisms for deadlock handling



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Memory Management

- ❑ All **data** must be in memory before and after processing
- ❑ All **instructions** must be in memory in order to execute
- ❑ Memory management determines what is in memory
 - Optimizing CPU utilization and computer response to users
- ❑ Memory **management activities**
 - Allocating and de-allocating memory space as needed
 - Keeping track of which parts of memory are currently being used and by whom



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Storage Management

- ❑ OS provides **uniform, logical** view of information storage
 - Each medium is controlled by device (i.e., disk drive, tape drive)
 - Varying properties include access speed, capacity, data-transfer rate, access method (sequential or random)
 - Abstracts physical properties to logical storage unit – *file*



Storage Management

□ File-System management

- Files usually organized into **directories**
- **Access control** on most systems to determine who can access what
- OS activities include
 - Creating and deleting files and directories
 - Primitives to manipulate files and directories
 - Mapping files onto secondary storage
 - Backup files onto stable (non-volatile) storage media



Mass-Storage Management

- ❑ Usually disks used to store data that does not fit in main memory or data that must be kept for a “**long**” period of time.
- ❑ **Entire speed** of computer operation hinges on disk subsystem and its algorithms
- ❑ OS activities
 - Free-space management
 - Storage allocation
 - Disk scheduling

依赖于



I/O Subsystem

- ❑ One purpose of OS is to **hide peculiarities** of hardware devices from the user
- ❑ I/O subsystem responsible for
 - Memory management of I/O including
 - **buffering** (storing data temporarily while it is being transferred),
 - **caching** (storing parts of data in faster storage for performance),
 - **spooling** (the overlapping of output of one job with input of other jobs)
 - General device-driver interface
 - Drivers for specific hardware devices



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Protection and Security

- ❑ **Protection** – any mechanism for **controlling access** of processes or users to resources defined by the OS
- ❑ Systems generally first distinguish among users, to determine **who can do what**
 - User identities (user IDs, security IDs) include name and associated number, one per user
 - User ID then associated with all files, processes of that user to determine access control
 - Group identifier (group ID) allows set of users to be defined and controls managed, then also associated with each process, file



Protection and Security

- ❑ **Security – defense** of the system against internal and external **attacks**
 - Huge range, including denial-of-service, worms, viruses, identity theft, theft of service



End of Chapter 1

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